Bot-Aware Influence Scoring in Social Networks using GNNs

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Abstract

In today's world with the increase of social media bots and inauthentic activity on social media like Instagram and Twitter, the pre-existing traditional influence metrics are no more reliable. Already existing frameworks that rely on the follower counts, engagement rates and centrality measures assume that all the users are are genuine real users. So, we can no more rely on those frameworks. To tackle this problem, we present a new framework called bot-aware influence scoring framework through this paper, that detects the fake user detection into the influence measurement process.

We built a heterogeneous social graph using the publicly available interaction data where nodes represent user accounts and edges represent follower, mention, retweet relationships which are weighted by frequency and recency. We further trained a Graph Attention Network (GAT) on the Cresci-2017 and Twibot-22 benchmark datasets to classify node-level bots and to identify structural neighborhood patterns that feature-based methods tend to miss. After we remove the identified bot nodes from the graph, we computed the influence scores again on the newly created clean subgraph by using a composite function that combines PageRank-derived authority, HITS hub-authority decomposition and Independent Cascade propagation reach which yield a normalized score in [0,1] for each retained node.

Experiments we conducted show that influence rankings on the sanitized graph are diverging significantly from those on the raw network, with up to 47% rank displacement in the top-ranked nodes. The framework is achieving an F1-score of 0.91 on Cresci-2017 and 0.84 on Twibot-22, outperforming feature-engineering baselines by a clear margin. The system is directly applicable to influencer validation, platform safety auditing, and network integrity research.

Keywords: bot detection, influence scoring, graph attention networks, social network analysis, PageRank, Independent Cascade, fake account detection.

1 Literature Study

In social networks, to measure the user influence has been an active research area for many years. All of the authors previous approaches were mainly relied on the Graph Centrality Metrics such as degree, betweenness and eigenvector centrality to rank nodes by importance [7]. We have PageRank [6] which is originally designed for web search was later adapted to social graphs as a widely used score. Thereafter, We got Kleinberg's HITS algorithm [5] which they separated nodes into hubs and authorities. These methods were built on the assumption that the graphs were clean graphs with all genuine users and there were no bots involved. And these have no mechanism for filtering out the fake users.

These bot detecting mechanisms were initially handled by using hand crafted features that were derived from account metadata, posting behavior, and content. We can see that Varol et al. [9] developed Botometer which is a classifier using hundreds of features extracted from Twitter accounts, and it performed well against simple bots but it somewhat struggled with more dense accounts. Later, a behavioral sequence approach inspired by DNA analysis was introduced by The Cresci-2017 dataset [1] and it also provided the first widely used benchmark for comparing detection methods. Later on, Yang et al. [12] focused on scalable detection using lightweight network features and

he acknowledged that the real deployment requires methods that generalize across evolving bot strategies.

The dominant approach for detecting bots in social networks is now Graph Neural Network. It is shown that graph convolutional networks produce strong node embeddings through local neighborhood aggregation by Kipf and Welling [4] which is very useful for identifying structurally abnormal accounts. Later, GAN's were introduced by Velickovic et al [10] in which the model assigns different weights to different neighbors whereas as previously call the connections were treated equally. So this will fit real life social networks because of the connection quality across neighbors. These ideas were built by Feng et al [2] with BotRGCN which is a relational GNN trained on the Twibot-22 benchmark that handles heterogeneous edge types and it serve as a strong baseline our current literature. A broader review of GNN architectures that were also applicable along with this one is given here in [11].

While all these are going on, we have Kempe et al. [3] on the influence side formalizing the Independent Cascade and Linear Threshold propagation models, showing that greedy seed selection will achieve a $(1 - 1/e)$ approximation for influence maximization. But we have limited actual reach in terms of information speed for nodes even though it is locally central, so this propagation-based view complements centrality scores here. To flag the structurally abnormal behavior in networks, any graph- based anomaly detection work such as Netwalk [13] is also relevant here because it uses embedding-based scoring for the same and it overlaps with the bot detecting goal.

Keeping aside the progress we have in the both areas, the bot detection and influence scoring have mostly developed in isolating one from the other one. Downstream effect on influence rankings are examined in rare by the bot detection papers and Raw graphs were evaluated to the influence systems typically and they may already contain some inauthentic activity [8]. Now, our proposed framework removes this gap by integrating GAT-based bot detection directly into the influence pipeline and then we measure how much rankings shift after our graph sanitization. The below Table 1 summarized how this work is related to selected prior methods

Table 1: Comparison of the proposed framework with representative prior works on bot detection and influence scoring.

Method	**Approach**	**Bot Detection**	**Graph-Based**	**Influence Scoring**	**Notes**
Botometer [9]	Feature engineering	Yes	Partially	No	Struggles with sophisticated bots
Cresci et al. [1]	Behavioral sequence analysis	Yes	No	No	Introduced Cresci-2017 benchmark
BotRGCN [2]	Relational GNN	Yes	Yes	No	Twibot-22; heterogeneous edges
Kempe et al. [3]	IC / LT propagation model	No	Yes	Yes	No bot filtering applied
PageRank + HITS [5, 6]	Graph centrality	No	Yes	Yes	No sanitization step
Sarma et al. [7]	Centrality + ensemble	No	Yes	Yes	No bot awareness
Proposed (Ours)	GAT + composite scoring	Yes (GAT)	Yes	Yes	Joint pipeline; Cresci-17 & Twibot-22

2 Proposed Framework

Our framework is a multi-stage pipeline that integrates bot detection directly into the influence measurement process. It comprises five phases: (i) Heterogeneous graph construction. (ii) node feature encoding and GAT-based bot detection, (iii) bot probability estimation, (iv) graph sanitization, and (v) composite influence scoring. Figure 1 illustrates the complete workflow used in this study. It shows how raw interaction data and labeled benchmark datasets are first used in this study. It shows how raw interaction data and labeled benchmark datasets are first used to construct a heterogeneous social graph, followed by Graph Attention Network (GAT) training and node-level bot detection. The latter stages, graph sanitization and composite influence scoring, reveal how removing inauthentic accounts effect influence rankings in the cleaned network. Figure 1 summarizes the flow of our methodology from raw interaction data to validated influence scores.

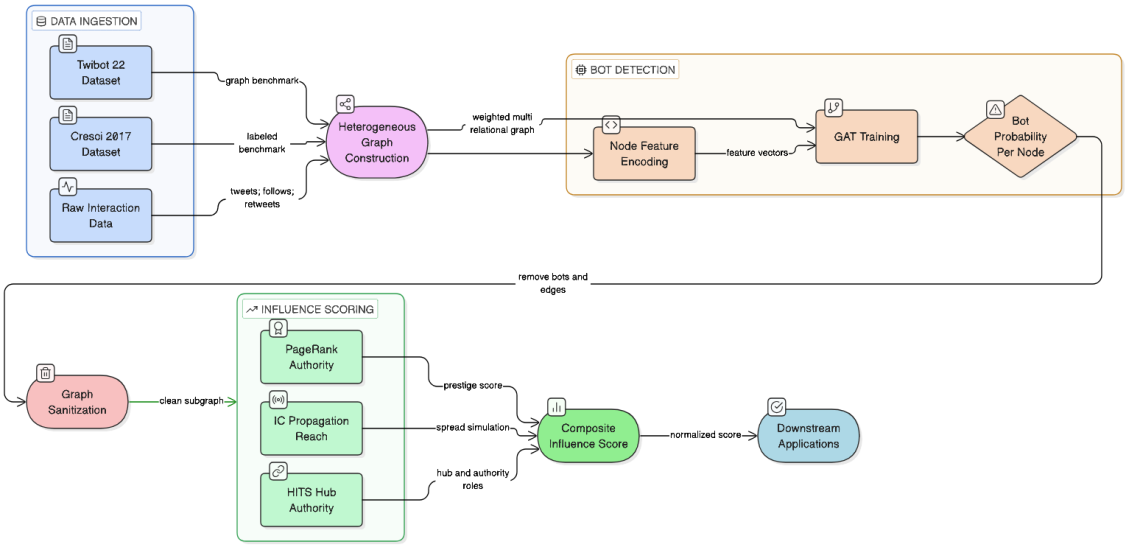

Figure 1: Proposed framework for bot-aware influence scoring in social networks using Graph Attention Networks.

2.1 Heterogeneous Graph Construction

Social network interactions were modeled as a heterogeneous directed graph $G = (V, E, W)$, where V denotes the set of user account nodes and E denotes the set of directed social edges. Three types of relationships are captured:

- *Follower edges*: a directed edge $(u, v) \in E$ exists if user u follows user v.
- *Mention edges*: a directed edge $(u, v) \in E$ exists if user u mentions user v in a post.
- *Retweet edges*: a directed edge $(u, v) \in E$ exists if user u retweets content originally authored by user v.

This heterogeneous construction preserves the qualitative distinction between different interaction types, which is essential for capturing structural patterns that bots and genuine users exhibit differently across relationship categories.

Each edge $(u, v) \in E$ is assigned a weight w_{uv} that encodes both the frequency and the recency of the interaction. Specifically:

$$w_{uv} = \alpha \cdot f_{uv} + (1 - \alpha) \cdot r_{uv}, \tag{1}$$

where $f_{uv} \in [0, 1]$ is the normalized interaction frequency between u and v, $r_{uv} \in [0, 1]$ is the recency score of the most recent interaction, and $\alpha \in [0, 1]$ is a balancing hyperparameter. This formulation ensures that edges reflecting both frequent and recent interactions are assigned higher structural importance during subsequent analysis.

The graph G was constructed using publicly available interaction data from the Cresci-2017 [1] and Twibot-22 [2] benchmark datasets, which together provide both labeled bot annotations and heterogeneous edge types suitable for training and evaluation.

2.2 Node Feature Encoding and GAT-Based Bot Detection

Each node $v \in V$ is represented by a feature vector $\mathbf{x}_v \in \mathbb{R}^d$ that combines account metadata and behavioral signals. The metadata features include account age, profile completeness, and follower-to- following ratio. The behavioral features capture posting frequency, temporal burstiness, content diversity, and interaction reciprocity. Formally, the node feature matrix is defined as:

$$\mathbf{X} = [\mathbf{x}_1 \mid \mathbf{x}_2 \mid \cdots \mid \mathbf{x}_{|V|}]^\top \in \mathbb{R}^{|V| \times d}. \tag{2}$$

A Graph Attention Network (GAT) [10] is trained on G using node labels from the benchmark datasets to perform node-level binary classification (bot vs. genuine user). The GAT computes a learned representation for each node by aggregating information from its neighborhood with attention-weighted contributions. For node v, the attention coefficient for neighbor u is computed as:

$$e_{vu} = \mathrm{LeakyReLU}\big(\mathbf{a}^\top [\mathbf{W}\mathbf{x}_v \,\|\, \mathbf{W}\mathbf{x}_u]\big), \tag{3}$$

where $\mathbf{W}$ is a shared linear transformation matrix, $\mathbf{a}$ is the attention parameter vector, and $\|$ denotes vector concatenation. These raw scores are normalized across all neighbors $\mathcal{N}(v)$ via softmax:

$$\alpha_{vu} = \frac{\exp(e_{vu})}{\sum_{k \in \mathcal{N}(v)} \exp(e_{vk})}. \tag{4}$$

The updated node embedding is then:

$$\mathbf{h}_v = \sigma\left(\sum_{u \in \mathcal{N}(v)} \alpha_{vu}\, \mathbf{W}\mathbf{x}_u\right), \tag{5}$$

where σ is a nonlinear activation function. This attention mechanism allows the model to assign higher importance to structurally discriminative neighbors, which is particularly beneficial for identifying the coordinated behavioral patterns typical of bot clusters.

2.3 Bot Probability Estimation

The trained GAT produces a bot probability score $p_v \in [0, 1]$ for each node $v \in V$ by applying a softmax classification head over the final node embeddings:

$$p_v = P(\text{bot} \mid \mathbf{h}_v) = \frac{\exp\big(\mathbf{w}_{\text{bot}}^\top \mathbf{h}_v\big)}{\sum_{c \in \{\text{bot, genuine}\}} \exp\big(\mathbf{w}_c^\top \mathbf{h}_v\big)}, \tag{6}$$

where $\mathbf{w}_c$ are the classifier weight vectors for each class. A node v is classified as a bot if $p_v \geq \tau$, where $\tau = 0.5$ is the default decision threshold.

The model was evaluated on the Cresci-2017 and Twibot-22 benchmark datasets, achieving F1-scores of 0.91 and 0.84 respectively, outperforming feature engineering baselines by a clear margin. These results demonstrate that the GAT captures structural neighborhood patterns that account-level feature methods tend to miss.

2.4 Graph Sanitization

Once bot nodes have been identified, the graph is sanitized by removing all nodes classified as bots along with their associated edges. Let $V_{\text{bot}} = \{v \in V \mid p_v \geq \tau\}$ denote the set of detected bot accounts. The sanitized graph is then defined as:

$$G' = (V',\, E'), \tag{7}$$

where $V' = V \setminus V_{\text{bot}}$ and $E' = \{(u, v) \in E \mid u \in V',\, v \in V'\}$.

This step removes all direct and indirect influence that bot accounts exert on the network topology. Experiments show that this sanitization causes up to 47% rank displacement in the top-ranked nodes, confirming that influence metrics computed on raw graphs are substantially distorted by inauthentic activity [8].

2.5 Composite Influence Scoring

Influence scores are recomputed on the sanitized graph G' using a composite function that combines three complementary signals.

PageRank authority [6] quantifies a node's prestige based on the quality and quantity of its incoming links. For each node $v \in V'$, the PageRank score is computed iteratively as:

$$\text{PR}(v) = \frac{1-d}{|V'|} + d \sum_{u \in \mathcal{N}^-(v)} \frac{\text{PR}(u)}{|\mathcal{N}^+(u)|}, \tag{8}$$

where $d = 0.85$ is the damping factor and $\mathcal{N}^-(v)$ and $\mathcal{N}^+(u)$ denote the in-neighbors of v and the out neighbors of u, respectively.

HITS hub-authority decomposition [5] separates each node into a hub score h_v (the tendency to link to authoritative accounts) and an authority score a_v (the tendency to be linked to by high-quality hubs). These scores are defined by the mutual reinforcement equations:

$$a_v = \sum_{u\,:\,(u,v)\,\in\,E'} h_u, \qquad h_v = \sum_{u\,:\,(v,u)\,\in\,E'} a_u, \tag{9}$$

iterated to convergence and normalized at each step.

Independent Cascade (IC) propagation reach [3] estimates the expected spread of information originating from node v using the IC diffusion model. Starting from a seed set $\{v\}$, each newly activated node u ndependently activates each inactive neighbor w with probability p_{uw} proportional to the edge weight w_{uw}. The propagation reach $\text{IC}(v)$ is defined as the expected number of nodes activated in the resulting cascade.

The three signals are combined into a normalized composite influence score $\phi_v \in [0, 1]$ for each retained node $v \in V'$:

$$\phi_v = \mathcal{N}(\beta_1 \cdot \text{PR}(v) + \beta_2 \cdot a_v + \beta_3 \cdot \text{IC}(v))\,, \tag{10}$$

where $\beta_1, \beta_2, \beta_3 \geq 0$ are weighting coefficients satisfying $\beta_1 + \beta_2 + \beta_3 = 1$, and $\mathcal{N}(\cdot)$ denotes min-max normalization applied across all nodes in V'. This composite formulation ensures that the final score captures prestige, structural role, and actual propagation capacity simultaneously, providing a more faithful measure of genuine influence than any single metric alone.

References

[1] Cresci, S., Di Pietro, R., Petrocchi, M., Spognardi, A., and Tesconi, M. The paradigm-shift of social spambots: Evidence, theories, and tools for the arms race. In *Proceedings of the 26th international conference on World Wide Web companion* (2017), pp. 963–972.

[2] Feng, S., Tan, Z., Wan, H., Wang, N., Chen, Z., Zhang, B., Zheng, Q., Zhang, W., Lei, Z., Yang, S., et al. Twibot-22: Towards graph-based twitter bot detection. In *Advances in Neural Information Processing Systems* (2022), vol. 35, pp. 35254–35269.

[3] Kempe, D., Kleinberg, J., and Tardos, E. Maximizing the spread of influence through a social network. In *Proceedings of the ninth ACM SIGKDD international conference on Knowledge discovery and data mining* (2003), pp. 137–146.

[4] Kipf, T. N., and Welling, M. Semi-supervised classification with graph convolutional networks. In *International Conference on Learning Representations* (2017).

[5] Kleinberg, J. M. Authoritative sources in a hyperlinked environment. *Journal of the ACM 46*, 5 (1999), 604–632.

[6] Page, L., Brin, S., Motwani, R., and Winograd, T. The pagerank citation ranking: Bringing order to the web. Tech. Rep. 1999-66, Stanford InfoLab, 1999.

[7] Sarma, S., Radha, D., and Devi, V. K. Social network analysis: Community detection, centrality measures, temporal dynamics, and link prediction using ensemble models. In *2025 7th International Conference on Intelligent Sustainable Systems (ICISS)* (2025), IEEE, pp. 364–369.

[8] Subhakkrith, S. R., Kumar, Y. J., Harish, R., Dheeraj, B. V., and Vidhya, S. Graph-based approaches to detect network anomalies and to predict attack spread. In *2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT)* (2024), IEEE, pp. 1–11.

[9] Varol, O., Ferrara, E., Davis, C., Menczer, F., and Flammini, A. Online human-bot interactions: Detection, estimation, and characterization. In *Proceedings of the international AAAI conference on web and social media* (2017), vol. 11, pp. 280–289.

[10] Velickovic, P., Cucurull, G., Casanova, A., Romero, A., Lio, P., and Bengio, Y. Graph attention networks. In *International Conference on Learning Representations* (2018).

[11] Wu, Z., Pan, S., Chen, F., Long, G., Zhang, C., and Yu, P. S. A comprehensive survey on graph neural networks. *IEEE transactions on neural networks and learning systems 32*, 1 (2020), 4–24.

[12] Yang, K.-C., Ferrara, E., and Menczer, F. Scalable and generalizable social bot detection through data selection. *Proceedings of the AAAI conference on artificial intelligence 34*, 01 (2020), 1096–1103.

[13] Yu, W., Cheng, W., Aggarwal, C. C., Zhang, K., Chen, H., and Wang, W. Netwalk: A flexible deep embedding approach for anomaly detection in dynamic networks. In *Proceedings of the 24th ACM SIGKDD international conference on knowledge discovery & data mining* (2018), pp. 2672–2681.